10 Coordinate geometry: vectors and lines, a synthesis

Sheet 56 Tue 30 Mar. 2027 *Vectors and collinearity: consolidation*

1. Warm-up · Quick $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.
2. Core · Method Are the points $A(-1,2)$, $B(1,3)$ and $C(5,5)$ collinear?
3. Core + · Method $A(-1,2)$, $B(3,1)$, $C(4,4)$. Find D such that $ABCD$ is a parallelogram.
4. Review · Quick In a class of 24 pupils, 18 have school lunch. What proportion is this, as a percentage?
5. Thread · Prove Show that $(n+1)^2-(n-1)^2=4n$ for every integer n. What can be concluded about the difference of the squares of two consecutive odd numbers (n even)?

Sheet 57 Thu 1st Apr. 2027 *Determinant and equations of lines: consolidation*

1. Warm-up · Quick What is the gradient of the line $3x+2y-6=0$?
2. Core · Method Equation of the line (AB) with $A(1,2)$ and $B(4,8)$.
3. Core + · Method Are the lines $2x-y+1=0$ and $4x-2y+5=0$ parallel, coincident or intersecting?
4. Review · Quick Calculate $10^{-2}\times 5\times 10^{4}$.
5. Thread · Method Solve in $\mathbb{R}$: $7-2(x-1)=3x-6$.

Sheet 58 Mon 5 Apr. 2027 *Plane geometry problems: choosing a method*

1. Warm-up · Quick Distance AB for $A(0,0)$ and $B(5,12)$?
2. Core · Method $A(0,0)$, $B(4,2)$, $C(-2,4)$. What type of triangle is ABC?
3. Core + · Method Give two methods to prove, using coordinates, that $ABCD$ is a parallelogram.
4. Review · Reading In a class of 30 pupils, a bar chart shows x pupils with the mark 12 and $2x$ pupils with the mark 14; the other marks account for 18 pupils. Find x.
5. Thread · Quick A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.

Sheet 59 Tue 6 Apr. 2027 *Situations from other subjects*

1. Warm-up · Quick A moving object has position $x(t)=3t+2$ (in m, t in s). Where is it at $t=4$?
2. Core · Method A second object follows $x_2(t)=5t-4$. At what time and where do the objects meet?
3. Core + · Quick A velocity vector has coordinates $\begin{pmatrix}3\\4\end{pmatrix}$ (in m/s). What is its magnitude?
4. Review · Quick Calculate $\frac{1}{2}-\frac{3}{5}$.
5. Thread · Write-up ABC is isosceles at A and H is the foot of the altitude from A. What can be stated about H?

Sheet 60 Thu 8 Apr. 2027 *Communicating a proof: implication, converse, equivalence*

1. Warm-up · Quick State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".
2. Core · True/false True or false? Justify. The previous converse is true.
3. Core + · Write-up Is the equivalence "$x^2=4 \Leftrightarrow x=2$" correct? Correct it.
4. Review · Quick Compare 2^{10} and 10^{3}.
5. Thread · Quick Between which consecutive integers does $\frac{7}{4}$ lie on a number line? Give its decimal

value.

Sheet 61 Mon 26 Apr. 2027 *Class discussion: comparing methods*

1. Warm-up · Quick Distance between $A(-2,1)$ and $B(1,5)$?
2. Core · Method Give three methods to prove that A, B, C are collinear.
3. Core + · Reflection To prove that two lines given by $y = m_1x + p_1$ and $y = m_2x + p_2$ are parallel, which method is quickest?
4. Review · Factorise Factorise $6x + 9$.
5. Thread · Method Write 360 as a product of prime factors. How many divisors does it have?